WEEK #6

1# A grading system compares three internal-assessment scores. Write a program to find the highest score among the three, using only if statements.

```
a = float(input("Enter 1st score : "))

b = float(input("Enter 2nd score : "))

c = float(input("Enter 3rd score : "))

if a < 0 or b < 0 or c < 0:

   print("Invalid score entered")

d = a

if b > a and b > c:

   d = b

if c > b and c > a:

   d = c

print("Highest score is: ", d)
```

```
Enter 1st score : 14
Enter 2nd score : 17
Enter 3rd score : 17.5
Highest score is:  17.5
```

2# A sensor module collects readings from three temperature probes. Write a program to determine the lowest reading using if–else logic.

```
a = float(input("Enter temperature 1: "))

b = float(input("Enter temperature 2: "))

c = float(input("Enter temperature 3: "))

if b < a:

   if b < c:

      d = b

else:

   if c < a:

      d = c

   else:

      d = a

if a < 0 or b < 0 or c < 0:

   print("Invalid score entered")
```

```
print("Lowest score is: ", d)
```

```
Enter temperature 1: 34
Enter temperature 2: 19
Enter temperature 3: 24
Lowest score is:  19.0
```

3# Write a program to check a year for leap year. (Use if and else)

```
year = int(input("Enter year: "))
if year > 0:
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        print("Leap Year")
    else:
        print("Not a Leap Year")
else:
    print("Not a valid year")
```

```
Enter year: 2022
Not a Leap Year
```

4# Write a program to print number of days in a month. (Chained Conditional)

```
month = int(input("Enter month (1-12): "))
if month >= 1 and month <= 12:
    year = int(input("Enter year: "))
    if year >= 0:
        m = [1, 3, 5, 7, 8, 10, 12]
        if month in m:
            print("31 days")

        elif month == 2:
            if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
                print("29 days")
            else:
                print("28 days")
        else:
            print("30 days")
```

```
    else:
        ("Not a valid year")
else:
    print("Not a valid month")
```

```
Enter month (1-12): 4
Enter year: 2025
30 days
```

5# Write a Python program that takes a month (1–12) and a year, uses a list of days in each month, and prints the correct number of days. Make sure of leap year for February.

```
month = int(input("Enter month (1-12): "))
year = int(input("Enter year: "))
days = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]
if month == 2:
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        print(29)
    else:
        print(28)
elif 1 <= month <= 12:
    print(days[month - 1])
else:
    print("Invalid month")
```

```
Enter month (1-12): 10
Enter year: 2026
31
```

6# An architectural software checks the type of triangular panels used in design. Write a program to: • Compute the area of a triangle using its three sides. • Print whether it is equilateral, isosceles, or scalene.

```
import math
print("Enter sides of triangle: ")
a = float(input("Enter 1st side : "))
b = float(input("Enter 2nd side : "))
c = float(input("Enter 3rd side : "))
if (a > 0 and b > 0 and c > 0):
```

```
    if (a+b > c and b+c > a and a+c > b):
        if (a == b or b == c or a==c):
            print("Isosceles Triangle")
        elif (a == b and b == c):
            print("Equilateral Triangle")
        else:
            print("Scalene Triangle")
        s = (a+b+c)/2
        area = math.sqrt(s*(s-a)*(s-b)*(s-c))
        print("Area is: ", area)
    else:
        print("These sides cant form a triangle")
else:
    print("Invalid length")
```

```
Enter sides of triangle:
Enter 1st side : 5
Enter 2nd side : 6
Enter 3rd side : 4
Scalene Triangle
Area is:  9.921567416492215
```

7# A device logs data indexes between two given IDs. Given two numbers r1 and r2 (r1 < r2): Write a program to create a list of integers from r1 to r2 (inclusive) using range().

```
r1 = int(input("Enter r1: "))
r2 = int(input("Enter r2: "))
numbers = list(range(r1, r2 + 1))
print(numbers)
```

```
Enter r1: 4
Enter r2: 7
[4, 5, 6, 7]
```

8# A parser receives a mixed list of characters, symbols, and integers. Write a program to extract and add only numeric items using isinstance(). Example 1: • Input: [1, 2.5, 3, -4] • Output: 0 Example 2: • Input: [10, -5, 20] • Output: 25

```
n = int(input("Enter the number of list elements: "))
l = []
```

```
for i in range(n):
    m = input(f"Enter list element {i+1}: ")
    if m.lstrip('-').isdigit():
        l.append(int(m))
    else:
        l.append(m)
print(l)
total = 0
for i in l:
    if isinstance(i, int):
        total += i
print(total)
```

```
Enter the number of list elements: 5
Enter list element 1: 1
Enter list element 2: 2
Enter list element 3: 3
Enter list element 4: 4
Enter list element 5: 5
[1, 2, 3, 4, 5]
15
```

9# A game engine verifies whether level IDs are arranged in a proper continuous sequence. Write a Python program to check whether a given list of numbers is consecutive. Example 1: • Input: [4, 2, 3, 5] • Output: True • Input: [1, 2, 2, 3] • Output: False

```
n = int(input("Enter the number of list elements: "))
l = []
for i in range(n):
    m = input(f"Enter list element {i+1}: ")
    l.append(int(m))
print(l)
l.sort()
print(l)
consecutive = True
for i in range(len(l) - 1):
    if l[i+1] - l[i] != 1:
```

```
    consecutive = False
    break
print(consecutive)
```

```
Enter the number of list elements: 3
Enter list element 1: 4
Enter list element 2: 7
Enter list element 3: 8
[4, 7, 8]
[4, 7, 8]
False
```

10# Given an integer n, return a string array answer (1-indexed) where: • answer[i] == "FizzBuzz" if i is divisible by 3 and 5. • answer[i] == "Fizz" if i is divisible by 3. • answer[i] == "Buzz" if i is divisible by 5. • answer[i] == i (as a string) if none of the above conditions are true. Example : Input: n = 15 Output: ["1","2","Fizz","4","Buzz","Fizz","7","8","Fizz","Buzz","11","Fizz","13"," 14","FizzBuzz"]

```
n = int(input("Enter n: "))
list = []
for i in range(1, n + 1):
  if i % 3 == 0 and i % 5 == 0:
    list.append("FizzBuzz")
  elif i % 3 == 0:
    list.append("Fizz")
  elif i % 5 == 0:
    list.append("Buzz")
  else:
    list.append(str(i))
print(list)
```

```
Enter n: 6
['1', '2', 'Fizz', '4', 'Buzz', 'Fizz']
```